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OLLSCOIL NA hÉIREANN MÁ NUAD
THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

Year 3

Autumn Repeat Exam

2015 - 2016

Quantum Mechanics I
MP363

Professor S. Hands, Professor D. M. Heffernan, Dr. P. Watts

Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. Let $|\phi_n\rangle$ be the n^{th} orthonormalised energy eigenstate of a particle of mass m in an infinite square well of width a , with corresponding energy eigenvalue

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2, \quad n = 1, 2, \dots$$

Suppose the particle is prepared so as to be in the state

$$|\psi\rangle = \sum_{n=1}^{\infty} \frac{\alpha}{n^2} |\phi_n\rangle,$$

where α is a positive real number.

- (a) Determine the value of α such that $|\psi\rangle$ is normalised.

[15 marks]

- (b) Find the expectation value of the Hamiltonian for this state.

[15 marks]

- (c) What is the probability that a measurement of the energy gives $2\pi^2 \hbar^2 / ma^2$, and if this energy is found, what state is the particle in immediately afterward?

[20 marks]

The following sums may be useful:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

2. Suppose A is an operator on a Hilbert space $\mathcal{H}$ with inner product $\langle | \rangle$ and $A^\dagger$ is its adjoint.

- (a) Explain what it means for an operator to be *self-adjoint* and show that, if A is self-adjoint, then

$$\langle \psi | A | \zeta \rangle = \langle \zeta | A | \psi \rangle^*$$

for any ψ and ζ in $\mathcal{H}$.

[10 marks]

- (b) Prove that all eigenvalues of a self-adjoint operator are real.

[10 marks]

- (c) Use the inner product

$$\langle \psi | \zeta \rangle = \int_{-\infty}^{\infty} \psi^*(x) \zeta(x) dx$$

to prove that the two operators

$$A = i \frac{d}{dx}, \quad B = \frac{d^2}{dx^2}$$

are both self-adjoint. State and justify the assumptions you make about the behaviour of any functions you might use in this proof.

[30 marks]

3. The energy eigenfunctions for the infinite square well are $\phi_n(t, x) = u_n(x)e^{-iE_n t/\hbar}$ for $n = 1, 2, \dots$, where

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2, \quad u_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) & 0 \leq x \leq a, \\ 0 & x < 0, x > a \end{cases}$$

Suppose the state of a particle in such a well is given by the wavefunction

$$\psi(t, x) = \frac{3i}{\sqrt{10}} \phi_2(t, x) + \frac{1}{\sqrt{10}} \phi_3(t, x).$$

- (a) If the energy of this state is measured, what are the possible results, and with what probabilities do they occur?

[10 marks]

- (b) Compute $\langle X \rangle$ and $\langle P \rangle$ as functions of time.

[30 marks]

- (c) Use your results from (b) to confirm that

$$\langle P \rangle = m \frac{d}{dt} \langle X \rangle.$$

[10 marks]